

```
.In getPowerSet("BC") returning ['BC', 'B', 'C', '']  
In getPowerSet("ABC") part 2, get sets with head 'A'  
In getPowerSet("ABC") New set ABC  
In getPowerSet("ABC") New set AB  
In getPowerSet("ABC") New set AC  
In getPowerSet("ABC") New set A  
In getPowerSet("ABC") returning ['ABC', 'AB', 'AC', 'A', 'BC', 'B', 'C', '']  
['ABC', 'AB', 'AC', 'A', 'BC', 'B', 'C', '']
```

The `getPowerSet()` function accepts a single argument: the string `chars`, which contains the characters of the original set. The base case occurs when `chars` is the blank string ❶, representing an empty set. Recall that the power set is the set of all subsets of the original set. Thus, the power set of the empty set is simply a set containing the empty set, since the empty set is the only subset of the empty set. This is why the base case returns `['']`.

The recursive case is split into two parts. The first part is acquiring the power set of the tail of `chars`. We'll use the leap-of-faith principle and just assume the call to `getPowerSet()` returns the power set of the tail correctly ❷, even though at this point we'd still be in the process of writing the code for `getPowerSet()`.

To form the complete power set of `chars`, the second part of the recursive case forms new sets by adding the head to each of the tail power sets ❸. Together with the sets from the first part, this forms the power set of `chars` to return at the end of the function ❹.